

Unleashing the power of N-BEATS on the S&P Green Bond Index



With the ongoing focus on ESG and data science, the IFoA's Data Science, Sustainability and Climate Change Working Party continues its theme of marrying these fields

specifically earmarked at raising capital for projects and activities with environmental benefits. The proceeds from green bond issuances are dedicated to a range of projects including to address climate change, promote renewable energy and other environmentally-friendly initiatives.

The S&P Green Bond Index has been developed specifically to meet the growing demand for transparent and credible green bond benchmarks. It is one of several green bond indices published by S&P. These indices serve as a market benchmark for the ever growing green bond universe. The S&P Green Bond Index includes bonds flagged as 'green' by the non-profit organisation Climate Bond Initiative. The index was developed by the S&P Dow Jones Indices and Infrastructure Credit Alpha Group LLC.

We build on our [initial publication](#), where we shared our time series analysis of the S&P Green Bond Index using neural networks. We continue down the univariate pathway, delving into a much more intricate neural network design: N-BEATS.

Published in 2020, N-BEATS (Neural Basis Expansion Analysis Time Series) is a state-of-the-art time series forecasting model which uses a complex neural network architecture for accurate predictions.

It is a renowned deep learning approach, due to its unique design and consistent performance. In 2020 the original authors of this architecture demonstrated outperformance “improving forecast accuracy by 11% over a statistical benchmark and by 3% over last year’s winner of the M4 competition”.

For more details on N-BEATS, and our analysis, please see our report: [Time series analysis of GSS bonds: Part 2](#)

The code in our report is based on the Python-based Keras library and Google’s Tensorflow framework. The environment used for our analysis was Google Colab. For those interested, please see for example Google’s introductory tutorials via their website for Tensorflow (see link below). They also include tutorials on time series analysis with neural networks. Similar to our first paper, we have used the open-source library Optuna to assist with hyperparameter optimisation.

Background to working party

The Data Science, Sustainability and Climate Change Working Party, formed in 2022, is looking at a range of areas within the data science, sustainability, and climate change space. The working party is part of the lifelong learning pillar within the data science practice of the IFoA (please see below). We aim to add

Do keep an eye out for [future publications from our working party](#).

Background to data science lifelong learning

Lifelong learning is one of the 4 pillars within IFoA data science along with research, engagement, and professionalism and ethics. It is targeted at supporting development of actuaries within data science through a lens of 2 working parties:

- Membership and education: This party onboards members and external visitors to the world of data science via producing less technical but still valuable content such as articles, blog posts, and webinars. We also help the IFoA in enhancing its studying curve with modern data analytics approaches.
- Sustainability and climate change (see earlier)

Further reading

- [IFoA's lifelong learning data science page](#)
- [IFoA's data science practice area](#)
- [IFoA's data science research website](#)
- [IBM's introduction to data science](#)
- [Keras](#)
- [Optuna](#)
- [Tensorflow](#)

Share your views

What are your thoughts on the points raised in this article and paper? We would love to hear your views in the comments via the [IFoA's LinkedIn Data Science Group](#).



Authors



Debashish Dey

Co-chair, Climate Change Disclosures Working Party and Data Science, Sustainability and Climate Change Working Party; Chair, Unsupervised Learning Working Party until 6 May 2025



IFoA's Data Science, Sustainability and Climate Change Working Party

Popular topics

[AI](#)[Actuarial reports](#)[Annuities](#)[Banking](#)[Biodiversity](#)[CACT](#)[CMI](#)[Careers](#)[Chartered Actuary](#)[Climate Change](#)[Covid-19](#)[DEI](#)[IFoA Asia Conference](#)[IFoA Foundation](#)[IFoA conferences](#)[Neurodiversity](#)[Pensions](#)

All topics

[Contact Us](#)[Site map](#)[Visit Us](#)[Cookie Policy](#)[Accessibility Statement](#)[Working at the IFoA](#)[Terms and Conditions](#)[Find an Actuary](#)[The Actuary](#)



[The Actuary Jobs Asia](#)

[Privacy notice](#)

[Refund policy](#)

[Shop](#)

[CMI](#)



Copyright © Institute and Faculty of
Actuaries, 2014-2025.

